

SOFTWARE ENGINEERING

LAB3: COMPONENT MODELLING & ARCHITECTURAL PATTERN SELECTION

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1. Scenario: Digital Campus Library:

The Digital Campus Library is a system that allows students to access library services digitally. Students can search for books, check book availability, borrow and return books, and receive notifications. The system also provides authentication and manages library data through a centralized database.

2. Comparison of Architectural Patterns:

<i>Architectural Pattern</i>	<i>Description</i>	<i>Advantages</i>	<i>Disadvantages</i>
<i>Layered Architecture</i>	<i>The system is divided into layers such as presentation, business logic, and data access.</i>	<i>Easy to understand, develop, test, and maintain.</i>	<i>Changes may need to pass through multiple layers and can introduce additional processing.</i>
<i>Microservices Architecture</i>	<i>The system is divided into independent services, with each service handling a specific function.</i>	<i>Services can be developed, deployed, and scaled independently.</i>	<i>More complex to develop, deploy, and manage.</i>

<i>Architectural Pattern</i>	<i>Description</i>	<i>Advantages</i>	<i>Disadvantages</i>
<i>Client-Server Architecture</i>	<i>The client provides the user interface, while the server handles application logic and communicates with the database and other services.</i>	<i>Centralized management, easier access control, and suitable for multiple users accessing common library services.</i>	<i>The server can become a central point of failure if not properly designed.</i>

3. Selected Architectural Pattern:

Client-Server Architecture

Client-Server architecture is selected for the Digital Campus Library system.

Scenario-Specific Reasons

1. Centralized Library Management:

The library's book information, borrowing, returning, and availability operations can be managed centrally by the server, allowing multiple students to access the same updated information.

2. Multiple User Access:

Students can access the library services through a client interface while the server handles requests and communicates with the database and other services.

Security Advantage

Authentication and access control can be handled centrally by the server. This allows the system to verify users before providing access to library services and protects direct access to the database.

Performance Benefit

The server can centrally process database requests and library operations. This avoids requiring individual clients to directly communicate with the database and allows resources to be managed efficiently.

4. UML Component Diagram

Components Used

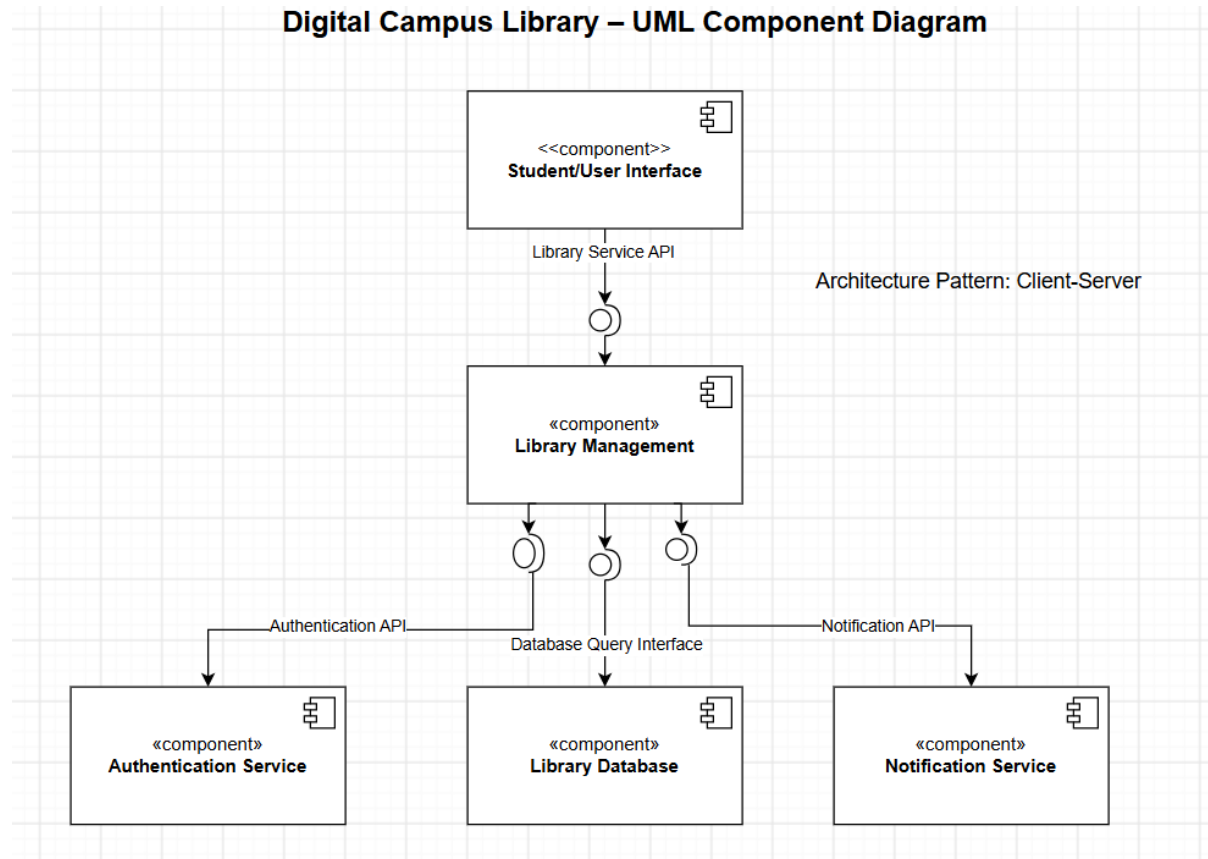
- 1. Student/User Interface*
- 2. Library Management*
- 3. Authentication Service*
- 4. Library Database*
- 5. Notification Service*

Interfaces Used

- 1. Library Service API*
- 2. Authentication API*
- 3. Database Query Interface*
- 4. Notification API*

The UML Component Diagram represents the relationships and dependencies between the components of the Digital Campus Library. The

required and provided interfaces show how components communicate with each other.



5. Conclusion

The Digital Campus Library was modelled using a UML Component Diagram with five major components and four interfaces. Three architectural patterns—Layered, Microservices, and Client-Server—were compared. Client-Server architecture was selected because it supports centralized library management, multiple user access, centralized authentication, and efficient communication between the application and database.

